

# **GENAI FOR DEVELOPERS**



 Lecture:  
APIs **vs** Local Models

## 1. Why This Matters

- In Generative AI development, you face a core decision:
  - → Should I call a cloud API like OpenAI or Gemini?
  - → Or should I run a local model like LLaMA or Mistral?
- This choice affects costs, performance, privacy, and deployment strategy.

## 2. Paid API Models: OpenAI (GPT-4o, GPT-3.5), Anthropic (Claude), Google Gemini.

- How they work:
  - Hosted by the provider (cloud).
  - You send a request via API → get back the model output.
  - Requires an API key for authentication.
- Pros:
  - State-of-the-art performance. No hardware requirements.
  - Fast to prototype (just an API call).
- Cons:
  - Ongoing cost per token. Rate limits on usage.
  - Data leaves your environment (privacy concern).

**3. Free / Local Models:** Hugging Face (open-source models like Falcon, Mistral), Ollama (easy local deployment), Meta's LLaMA family.

- **How they work:**
  - Model is downloaded and run locally on your machine/server.
  - Can be integrated with frameworks like Hugging Face transformers or Ollama CLI.
- **Pros:**
  - No per-token costs → free after setup.
  - Full control over model behavior.
  - Better data privacy (data never leaves machine).
- **Cons:**
  - Need powerful hardware (GPU/VRAM).
  - Models may be slower or less accurate than premium APIs.
  - More setup/configuration overhead.



# Trade-Offs at a Glance

| Factor   | Paid APIs (OpenAI, Anthropic, Gemini) | Local Models (Hugging Face, Ollama, LLaMA, Mistral) |
|----------|---------------------------------------|-----------------------------------------------------|
| Cost     | Pay-per-use (tokens)                  | Free (after setup) but hardware costs apply         |
| Speed    | Usually faster (optimized servers)    | Slower unless you have strong hardware              |
| Accuracy | SOTA models → very high               | Varies by model quality                             |
| Privacy  | Data leaves your environment          | Data stays local                                    |
| Control  | Limited (black box)                   | Full control, can fine-tune or modify               |
| Setup    | Easy (just API key)                   | Requires downloads, dependencies, config            |

# Setting Up Your Environment

- **For APIs:**
  - Install SDKs (openai, anthropic, google-generativeai).
  - Store API keys securely (.env, secrets manager).
- **For Hugging Face:**
  - Install transformers + huggingface\_hub.
  - Authenticate with a Hugging Face token.
- **For Ollama:**
  - Download and install Ollama (works on Mac/Linux, Windows preview).
  - Run ollama pull mistral to get models.
  - Query models locally with simple commands.

Code Demo 🙋



